Q] 4 memory hierarchy question -

① ~~what~~ where can a block be placed in the upper level

② How is a block found if it is in upper level

③ which block should be replaced on a miss.

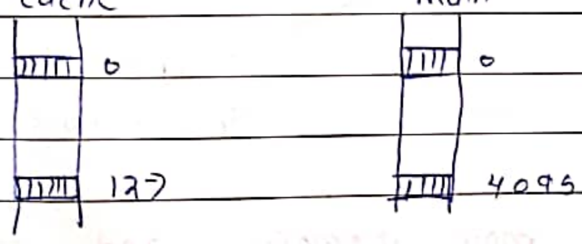
④ what happens on write.

For example, assume config. of, we have cache memory of 128 blocks and main memory of 4096 blocks, total words in a cache memory = 16 and main memory is also = 16

2k words   64k words $2^6 \times 2^{10} = 2^{16}$

2048 words   65,536 word   addresses can be issued

cache   main



① Ans ⇒ There's a ~~fo.~~ mechanism :-

Mapping functions :-

① Direct Mapping

x % size of cache

0 % 128

implementation is easy because of modulo division operator

Disadvantage ⇒ contention may arise.
{ over-writting & no utilisation of empty cache }

← 16 →

5	7	4
Tag	Block	word

③ Associative Mapping

* No contention / flexibility
* uses replacement algorithm
  Least recently used algorithm.

Disadvantag ⇒ complex implementation

first come first serve
random algorithm

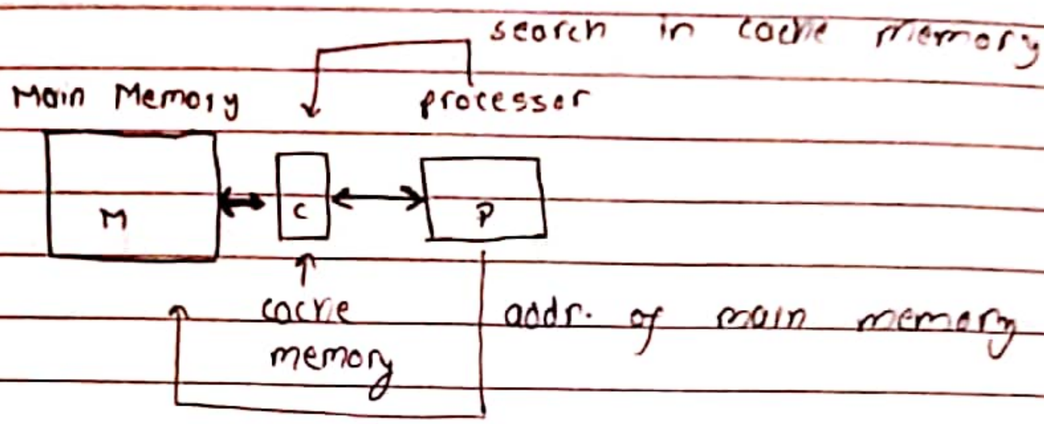
← 16 →

4096	12	4
	Tag	word

③ Set - Associative Mapping

combination of above two.

Cache Memory



* cache memory works on the principle of locality of reference.

* frequently used data in cache memory

* 90% time of a program
  10% portion of a program

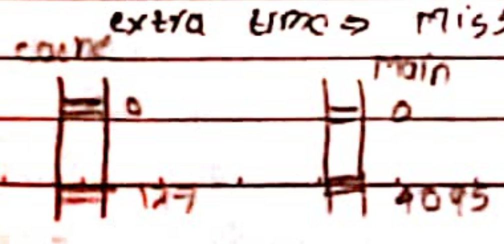
Locality of reference

① temporal — In terms of time

spatial — In terms of space

* Data transfer betⁿ main memory and cache is via **block**.

cache Hit ⇒ block available cache.
cache miss ⇒ block not ava. in cache
⇓
extra time ⇒ miss penalty



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512

Sector :- (5 ~~to~~ ~~77~~ byte)
one of the segment that make up a track
A sector is a smaller

seek :-
The process of ~~the~~ positioning read/write head over a proper track on

read/write head :-
HD uses a read/write head to access the disk.

platters
each have two recordable surface

Rotational speed
stack of platters is rotated at 5400 to 1500 RPM

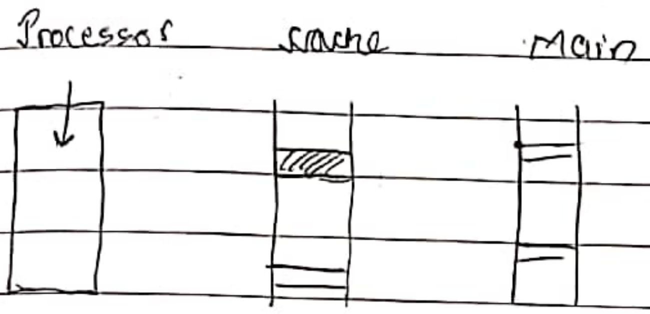
Diameter
1 - 3.5 Inches

No. of tracks per surface → 10k to 50k
innovation → 50 lakh
each track have 100 - 500 sectors

← 16 →

6	6	4
tag	set	word

Q) what happens on a write :-



Read hit ::

1.) write through → simultaneous updation in main memory
2.) write back ↓

~~each~~ cache memory में updation ~~करें~~ Eg 3
Eg 3(1) but not in main memory and assigns dirty bit indicating main memory में updation करें Eg 3(1), next time while over-writting updation will be done on main memory.

A Average Memory Access time for com. sys.

$$t = hC + (1-h)M$$

where h = hit rate
M = miss penalty
C = time to access info. in the cache

$(1-h) \Rightarrow$ miss rate

contains freq. accessed pages

TLB → Translation look aside buffer

Virtual address

TLB Hit

↓ yes

Read physical Add.

↓ No (Miss)

Page table Reading

Present

Absent

check in cache

Fault

↓

Read from storage

Hit

Miss

↓

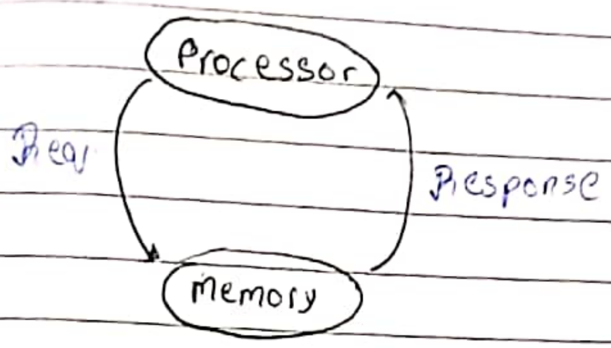
↓

Access data

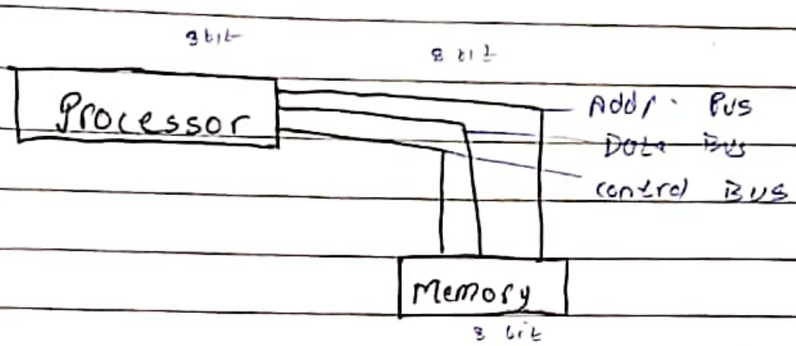
Access from main memory if page is present

6.) How many times the processor is referring to memory

Processor



I/O को अलग memory होती है। (extended memory)

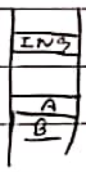


अलग sizes अलग है तो टुकड़ों में data transfer होगा

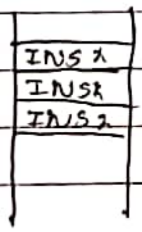
Add A, B

Instr. cycle (3 phases)

Fetch - Decode - Execute
↓
less amount of compared with fetch & execute, ∴ Ignore.



one instr. storing 3 memory locations



Fetch | Execute

machine cycle / Bus operation

Basic steps / RTL
Register transfer level activities / states
(S in Basic perf. equation)

Logical cylinder
All the tracks under the head at the given point on all the surface
(= no. of tracks)
To access data the OS must direct the disk through a three stage process.
1-> position the head over the proper track, this op- is called seek and the time to move the head to desired track is called the seek time. The disk manufacturer report min. seek time, max seek time and average seek time (avg => 3 milli sec to 30 milli sec).

2-> once the head reached the correct track, we ~~the~~ must wait for desired sector to rotate under the read write head.
Time => rotational delay / latency.
The average latency to desired is half away around the disk
(divide by 0.5)

3> Transfer time
Time to transfer a block of bits. It is a fn of sector size, rotational speed and recording density of the track

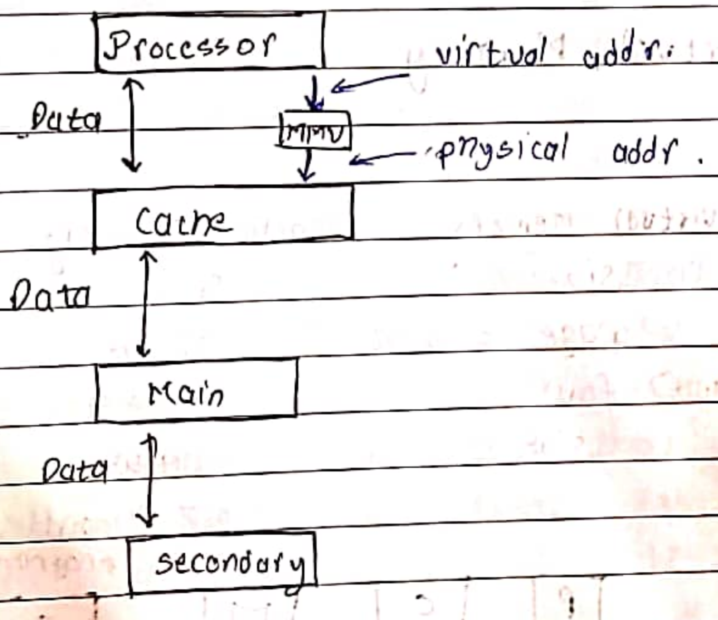
OS => swap in / swap out by DMA (Part of I/O) technique
Direct Memory access.

processor issues virtual OR logical Address.

virtual OR logical addr.
Translate ⇓ MMU
physical address.

Not converted x, translated

MMU => Memory management unit does the translation
→ combination of H/W & S/W.



$G = 2^{30}$, M

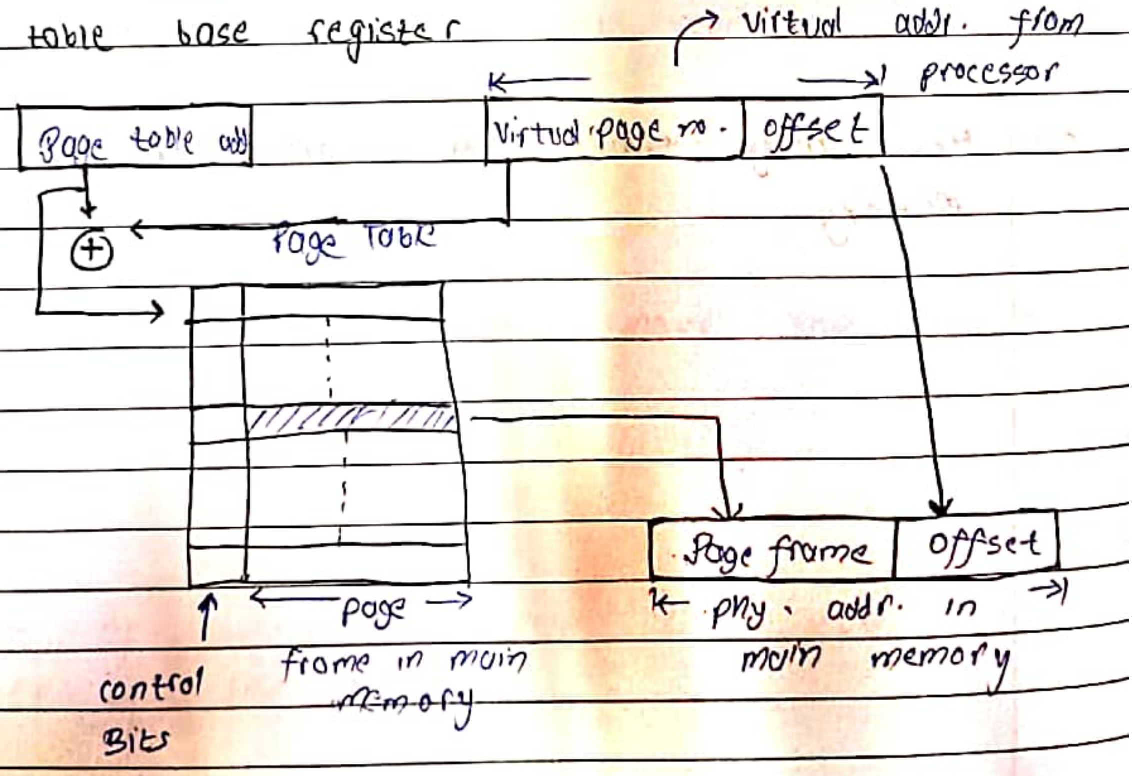
32 bit address , Addr. space ? $2^{32} = 4GB$
$2^{33} = 8GB$

only एक Bit बढ़ाने पर size required is just getting double ⇒ inefficient

* small pages में number of pages बढ़ जाएंगे or time के factors add-on होंगे।

* If large pages are used then memory waste होगी।

Addr. translation in virtual memory :-



hit time of L1 is one clock cycle, what is the average memory access time ?

40 ~~सं~~ ~~सं~~ 20 ~~सं~~

Ans:> $1 + 4\%(10 + 50\% \times 100)$

$1 + 0.04\ (10 + 50) \Rightarrow$ ~~$1 + 0.04 \times 60$~~

~~$1 + 75 = 76$~~

$1 + \frac{4}{100}[10 + 50] \Rightarrow 1 + \frac{4}{10\cancel{0}} \times 6\cancel{0}$

$\frac{10 + 24}{10} = \boxed{3.4 \text{ sec}}$

Q.] Assume a computer with no cache memory, it takes 10 clock cycles for each memory read access. Another computer system with cache memory needed 17 clock cycles to load a block into the cache. Ass. that 30% of the instructions in a typical prog. perform a read or write operation which means that there are 130 mem. access for every 100 instructions executed, ass. that the hit rates in the cache are 0.95 for instructions and 0.9 for data, let us further assume that the miss penalty is the same for both read and write access. Calc. the performance for^n these computers. $\boxed{C = 1}$

Sol:>

~~with cache~~ :-

$$\frac{\text{without cache}}{\text{with cache}} \Rightarrow \frac{130 \times 10}{100(0.95 \times 10 + 0.05 \times 17) + 30(0.9 \times 11 + 0.1 \times 17)}$$

instruction — read and write

$$\frac{1300}{100(180 + 78)} \Rightarrow \frac{1300}{258} = 5.0$$

* so computer system with cache is ≈ 5× times faster.

Q.] same que with ideal cache in 2nd PC.

sol>

$$\frac{130 \times 10}{100(1) + 30(1)} = \frac{130 \times 10}{130} = 10$$

$$\frac{258}{130} = 1.98$$

Virtual Memory

Virtual memory	cache memory
size	speed
page (2KB - 16KB)	Block
(page fault) fault	Miss
S/W	H/W

2 GB → main
2 GB ⇒ program

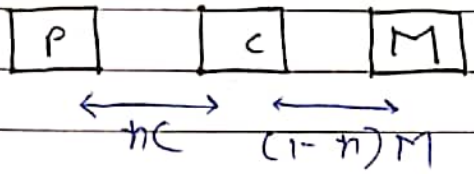
P | C | M | S

C ↓ portion of active program

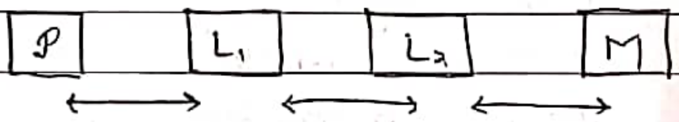
M ↓ Active program

$$h = \frac{\text{no. of hits}}{\text{total number of attempts}} \times 100\ \%$$

$$(1-h)\ (\text{miss rate}) = \frac{\text{no. of miss}}{\text{total number of attempts}} \times 100\ \%$$



* Levels of cache memory



t = hit time of L_1 + miss rate of L_1 * miss penalty of L_1

miss penalty of L_1 = hit time of L_2 + miss rate of L_2 * miss penalty of L_2

Q.] Suppose that in 1000 memory references, there are 40 in the 1st level cache and 20 misses in 2nd level cache, Assume the miss penalty of L2 cache to memory is 100 clock cycle, the hit time of L2 cache is 10 clock cycles, the

$1 \times 50 + 40 (10 + 100 \times 20) =$

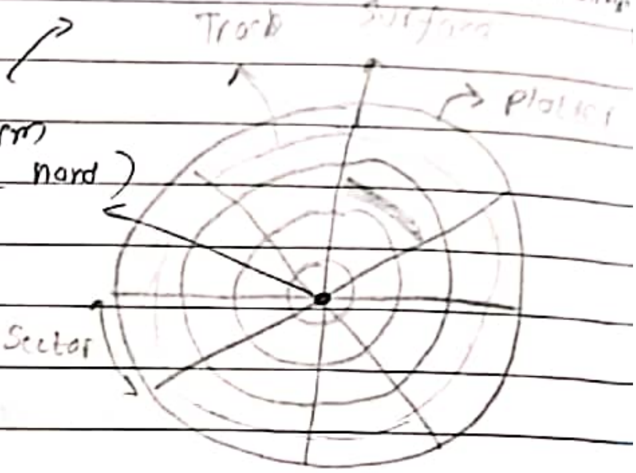
Secondary Storage :-

- Magnetic Disk
* sensitive
* made up of platters
  surface = 2 * platters

movement करेगा (track पर आगा)

* Track

arm
(~~sep read write~~ head)



Disk → collection of platters

Disk Drive → movement
rotation ... etc

Disk controller → info read and write rate ....

Logical cylinder → To store the program
→ reduces time to read and write

seek

seek time ⇒ arm / read-write head को particular track पर आने का time.

Rotational Delay → Time to move the track by rotation

Track

Thousands of concentric circles that makes up the surface of the disk

Problem exercise

Q7 What is the avg time to read and write a 512 byte sector for a typical disk rotating at 15000 RPM. The adverdised avg seek time is 4×10^{-3} s. (milli) The transfer rate is 100 MB per sec. and the controller over head is 0.2 mili sec, assume that the disk is Idle so that there is no waiting time. Avg disk acc. time?

Sol) Avg. disk access time = seek time + rotational + transfer + controller over head + waiting time

$$= 4 \times 10^{-3} + \frac{0.5}{15\ RPM} \times 10^{-3} + \frac{0.5\ KB}{100\ MB/s}$$

$$+ 0.2 \times 10^{-3} + 0$$

$$= 10^{-3} (4 + \cancel{0.033}\ 2 + \qquad + 0.2)$$

The first disk drive that have 1 GB capacity was produced by IBM in 1980. It was the size of kitchen appliances and weight 250 kgs and cost at that time 40,000 \$